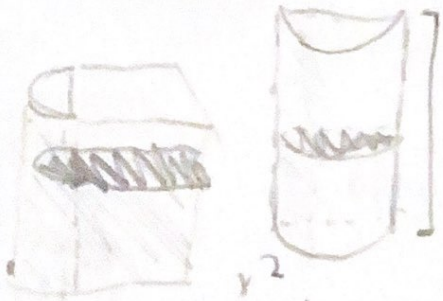
C14

(A)



200m

$F = \int_a^b \rho \cdot g \,(\text{length of strip})\,(\text{depth of strip})\,dy$

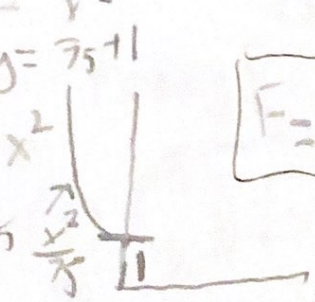
$F = \rho \cdot g \int_a^b (300 - y)\left(2\sqrt{75y - 75}\right) dy$

$F = 9{,}800 \int_1^{201} (201 - y)\left(2\sqrt{75y - 75}\right) dy$

$= 2x$ with $y = \frac{x^2}{75} + 1$

$75y - 75 = x^2$

$x = \sqrt{75y - 75}$



(B)

$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$

$2x = 8\sqrt{1 - y^2}$

Set $x = 0$:

$0^2 + \left(\frac{y}{3}\right)^2 = 1$

$\frac{y}{3} = \pm 1 \rightarrow y = -3, 3$

$3 = 3\sin y$

$1 = \sin y$

$\sin^{-1}(1) = \frac{\pi}{2}$

$F = \int_a^b \rho \cdot g \,(\text{length of strip})\,(\text{depth of strip})\,y\,dy$

$F = \int_{-3}^{3} \rho \cdot g \,(8\sqrt{1 - y^2})\,(3 - y)\,dy$

$F = 8\rho \cdot g \int_{-3}^{3} 3\sqrt{1 - y^2}\,dy$

$8\rho \cdot g \int_{-3}^{3} y\sqrt{1 - y^2}\,dy$

$8\rho \cdot g \int_{-3}^{3} 3\sqrt{1 - \frac{y^2}{9}}\,dy \rightarrow$ Trig sub: $3\sin(u)\,du$

$8\rho \cdot g \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 3\sqrt{1 - \sin^2(u)}\,\cos(u)\,du$

$24\rho \cdot g \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(u)\,du \rightarrow \frac{\cos(2u) + 1}{2}$

$12\rho \cdot g \left[\frac{1}{2}\sin(2u) + u\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$12\rho \cdot g \left[\left(\frac{1}{2}\sin(\pi) + \frac{\pi}{2}\right) - \left(\frac{1}{2}\sin(-\pi) - \frac{\pi}{2}\right)\right]$

$12\rho \cdot g \left[0 + \frac{\pi}{2} - 0 - \frac{\pi}{2}\right] = 24\pi \cdot \rho \cdot g - 8\rho \cdot g \int_{-3}^{3} 3\sin(u)\cos(u)$

$24\pi \cdot \rho \cdot g - 24\rho \cdot g \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin(u)\cos(u)\,du$

$x = 3\sin(u)$

$dx = 3\cos(u)$

B) i) $\left(\frac{x}{4}\right)^2+\left(\frac{y}{3}\right)^2=1$

$F=\rho\cdot g\int_{-3}^{3}\left(8\sqrt{1-\frac{y^2}{9}}\right)(3-y)\,dy$

$\frac{x}{4}=\sqrt{1-\frac{y^2}{9}}$

$F=8\rho\cdot g\left[\int_{-3}^{3}\left(\sqrt{1-\frac{y^2}{9}}\right)(3-y)\,dy\right]$

$x=4\sqrt{1-\frac{y^2}{9}}$

2x

Set $x=0$:

$\left(\frac{0}{4}\right)^2+\left(\frac{y}{3}\right)^2=1$

$\frac{y}{3}=\pm 1$

$y=-3,3$

$3=3\sin u$

$1=\sin u$

$\sin^{-1}(1)=\frac{\pi}{2},-\frac{\pi}{2}$

① $F=8\rho\cdot g\left[\int_{-3}^{3}3\sqrt{1-\frac{y^2}{9}}\,dy-\int_{-3}^{3}\sqrt{1-\frac{y^2}{9}}\,dy\right]$ ②

① $\int_{-3}^{3}3\sqrt{1-\frac{y^2}{9}}\cdot dy\rightarrow 3\int_{-3}^{3}\sqrt{1-\frac{y^2}{9}}\,dy$

Trig sub:

$\rightarrow y=3\sin u$

$dy=3\cos u$

$3\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sqrt{1-\frac{9\sin^2 u}{9}}\cdot 3\cos u\,du\rightarrow 3\int_{-3}^{3}\sqrt{1\cos^2 u}\cdot 3\cos u\,du$

$3\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}3\cos^2 u\,du\rightarrow 9\int_{-3}^{3}\frac{1+\cos(2u)}{2}\,du$ Half-angle

$\frac{9}{2}\left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}1\,du+\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos u\,du\right)\rightarrow\frac{9}{2}\left[[1]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}+[2\cos u]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\right]$

$\frac{9}{2}\left(\frac{\pi}{2}+\frac{\pi}{2}+0-0\right)\rightarrow\boxed{\frac{9\pi}{2}}$

② $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2(u)\cos^2(u)\,du\rightarrow$ u-sub: $v=\cos x$, $\cos\cdot\sin(x)dx$ $\rightarrow\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}-v^2\,dv\rightarrow$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}-\frac{\cos^3(x)}{3}+C$

$0-0=0$

$\frac{9\pi}{2}\cdot[8\rho\cdot g]\rightarrow\frac{9\pi}{2}[8\cdot 9\cdot 0.8]=997.518\ N$

ii) Total Force Be Greater than $\frac{1}{2}F$

$8\rho g\int_{-3}^{0}\sqrt{1-\frac{y^2}{9}}(3-y)\,dy\rightarrow$ ① $\frac{9}{2}\left[[1]_{-\frac{\pi}{2}}^{0}+[\cos u]_{-\frac{\pi}{2}}^{0}\right]\rightarrow\frac{9}{2}\left[-\frac{\pi}{2}+1\right]$

$-\frac{9\pi}{4}+\frac{9}{2}$

② $\int_{-\frac{\pi}{2}}^{0}-\frac{\cos^3(x)}{3}\rightarrow-\frac{1}{3}$

$8 \cdot P \cdot g \left[+ \frac{9\pi}{4} + \frac{9}{2} + \frac{1}{3} \right]$

$8 \cdot 9\text{Co} \cdot 9.8 \left[+ \frac{9\pi}{4} + \frac{29}{6} \right] = 839,799\ N$